

The Role of AI

Algorithm to bedside symposia





DATA STORIES

From algorithm to bedside

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03-14 Feb. 2025

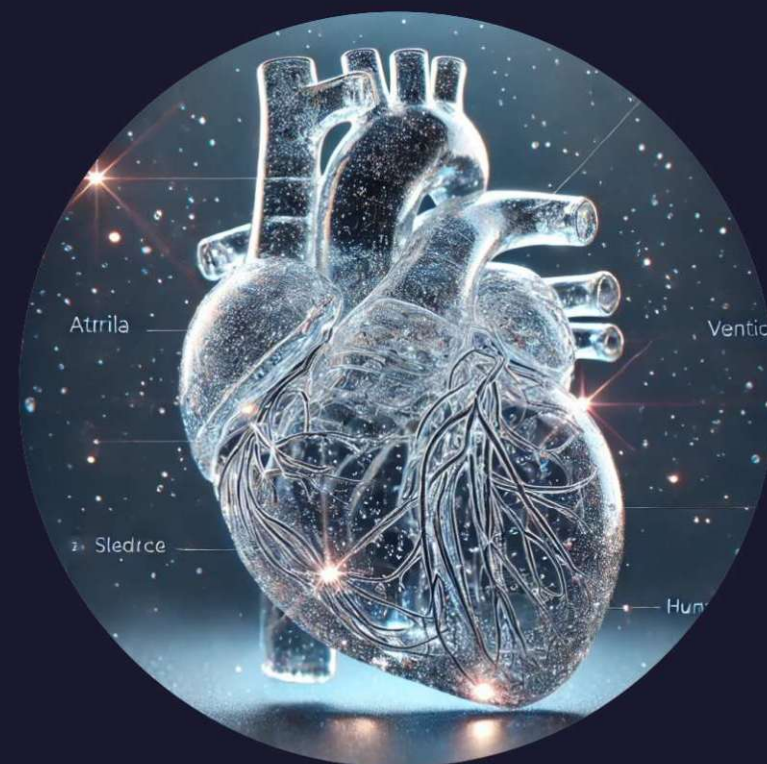
Lunch symposia

Omics-Based Clinical and Population Studies (**session 1: Mon. 3 Feb. 2025** - Specific Case Studies)

Contextual Epidemiology of Cardiometabolic Conditions, Polypharmacy and Multi-morbidity (**session 2: Thurs. 6 Feb. 2025**)

Scientific Methods in Biomedical Research (**sessions 3 and 4: Mon and Thurs. 10 and 13 Feb. 2025**)

- Omics in biomedical research
- Evidence pyramid and analytics
- Global Burden of Disease, Health Analytics, and Access to Therapeutics





Catch up

THERE ARE 3 KINDS OF LIES

lies, damned lies, and statistics



Antimicrobial Resistance : Global burden of AMR – Microbiome and Metagenomics

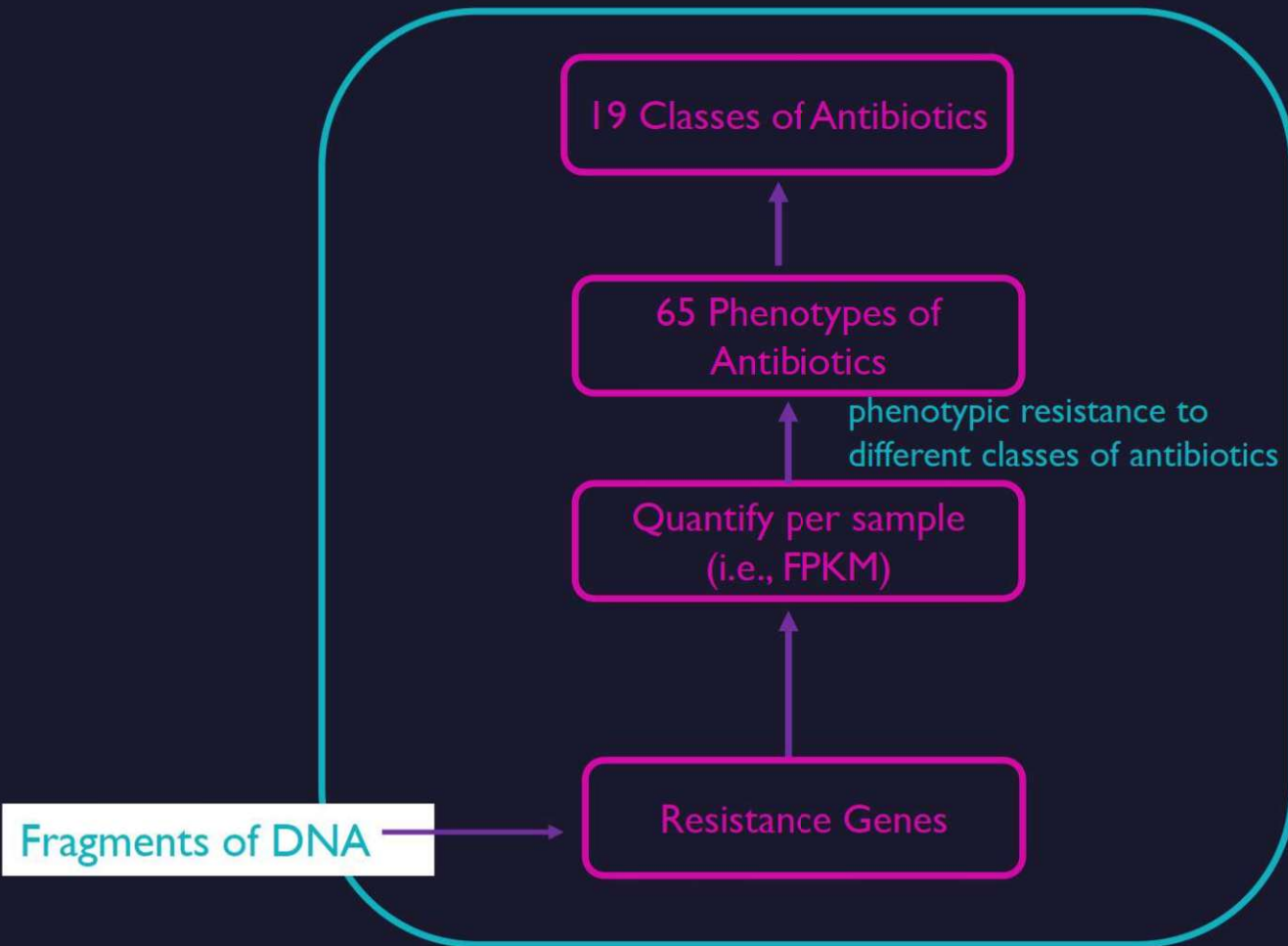


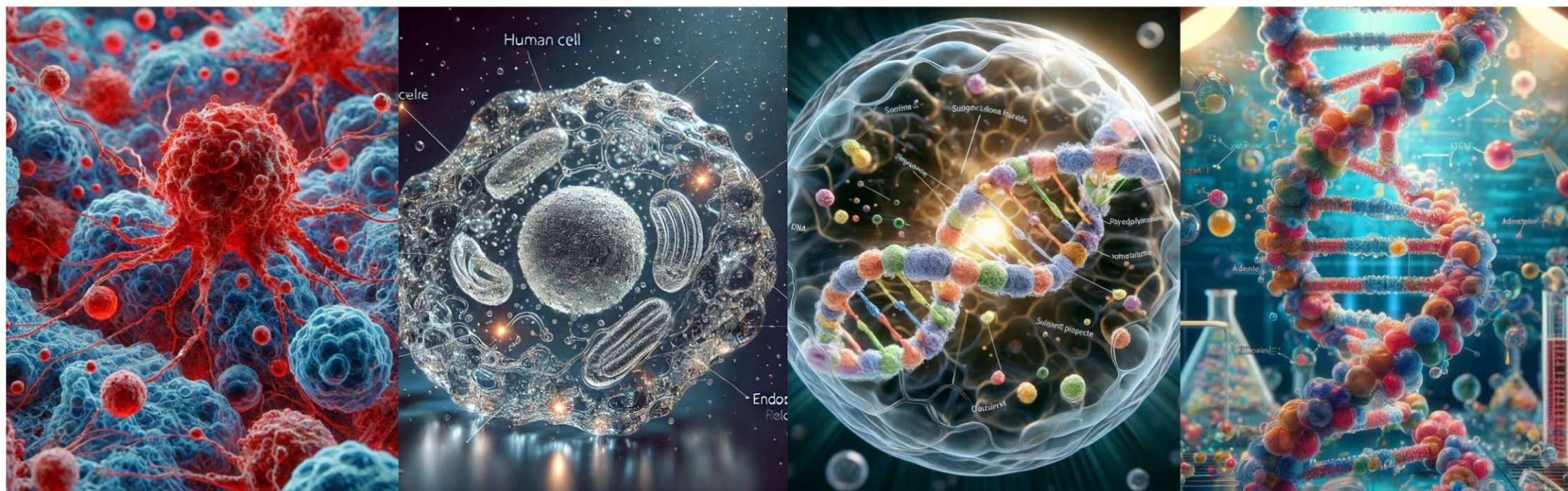
28-01-2025

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Own Project Pipeline





Cancer Epi-omics cohort

Epi-omics And Drug Use: Identification Of Late Effects Trajectories

28-01-2025

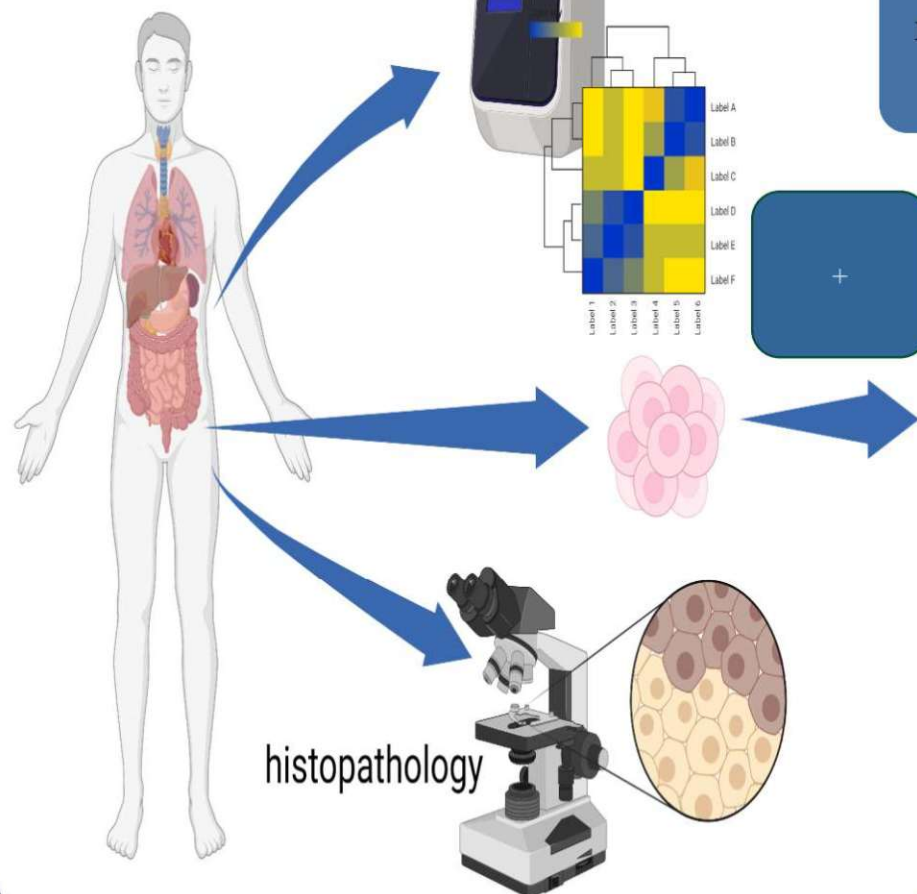
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Own data

NGS analysis



National cancer registries

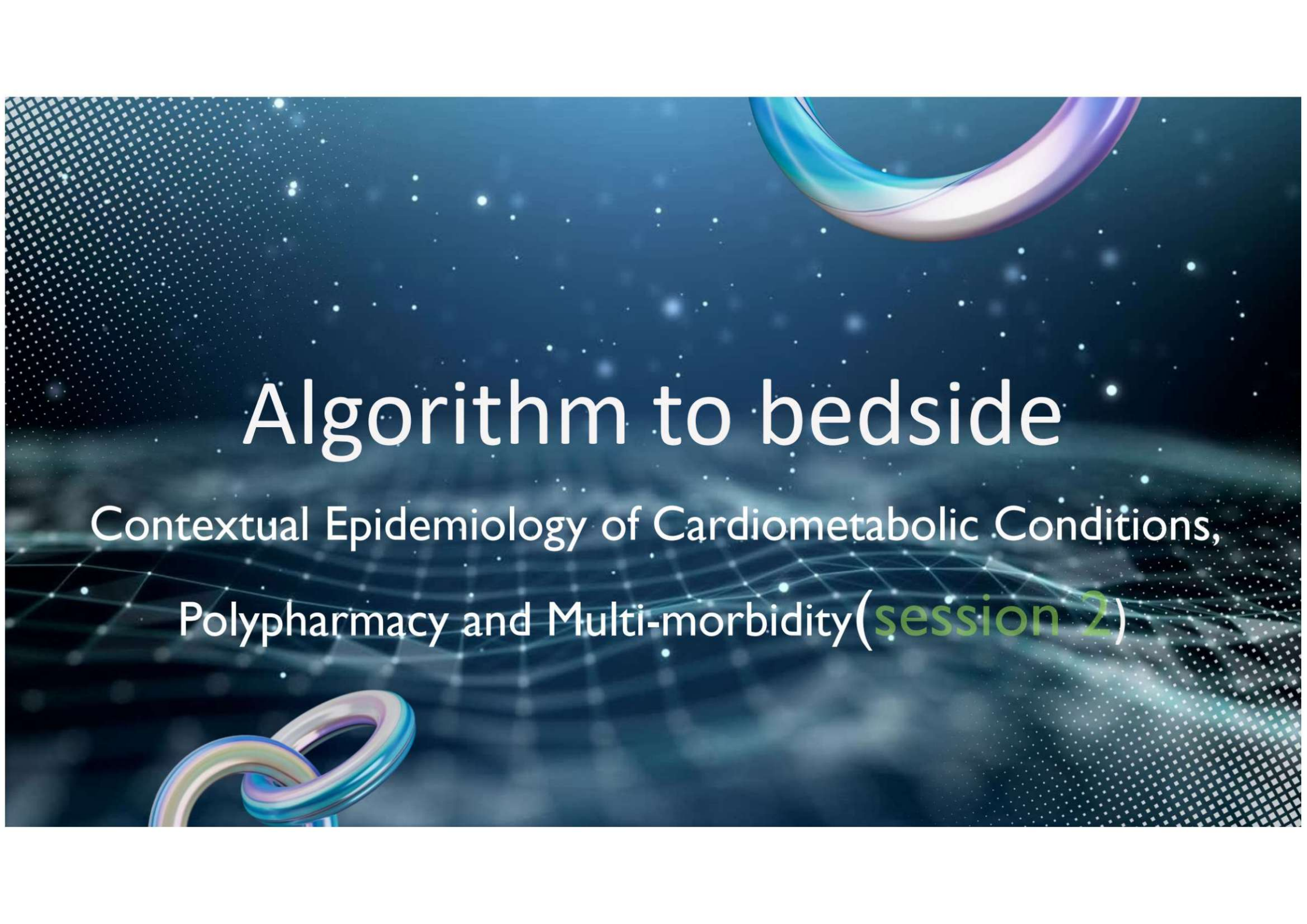
> 33,160
one-year
survivors

Matched by sex, age
and country

Central
population
registries

> 212,892
comparisons

National hospital
registries
(discharge
diagnoses according
to ICD to evaluate
late effects)

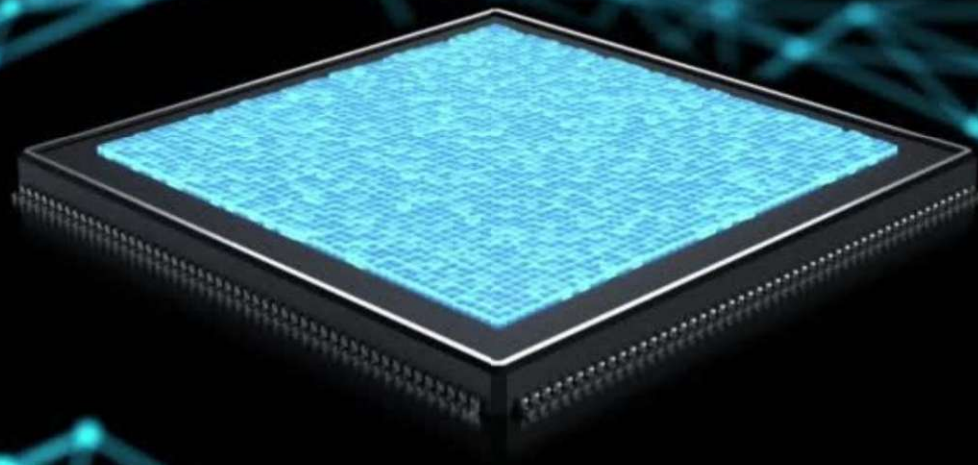


Algorithm to bedside

Contextual Epidemiology of Cardiometabolic Conditions,
Polypharmacy and Multi-morbidity(session 2)

Data with no context is noise

Algorithm to bedside

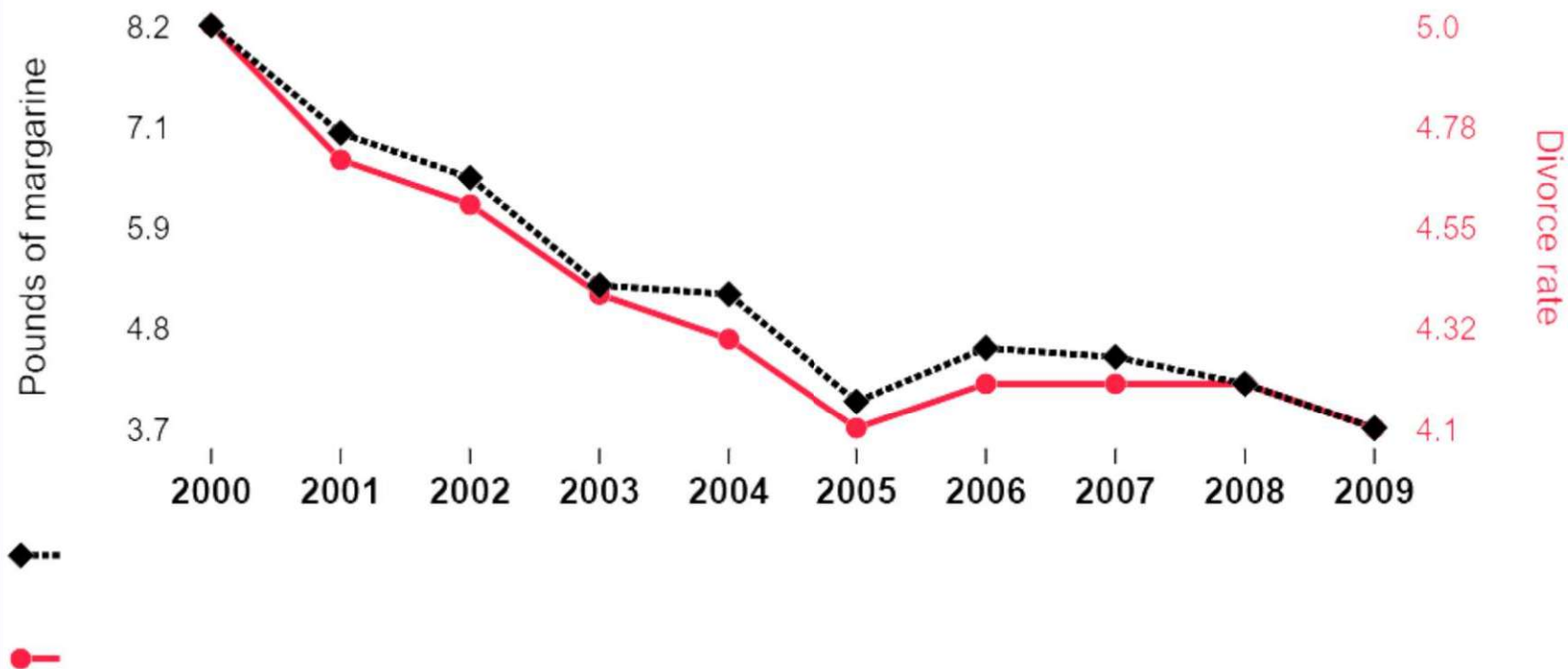




Per capita consumption of margarine

correlates with

The divorce rate in Maine



The background of the slide is a photograph of a historic brick building with a dark tiled roof. A clock is mounted on the wall between two windows. In the foreground, several people are riding bicycles through an arched gateway of the building. The scene is surrounded by lush green trees.

Patterns of Multimorbidity

2018-2019

KØBENHAVNS UNIVERSITET





2018-2019: Researcher, Research Unit of Chronic Conditions, CKFF

- Clusters of multimorbidity in the general population: a systematic **review**.
- Clusters of multimorbidity in the **Danish** population.
- Utilization of national **healthcare services** by individuals with multimorbidity in Denmark.



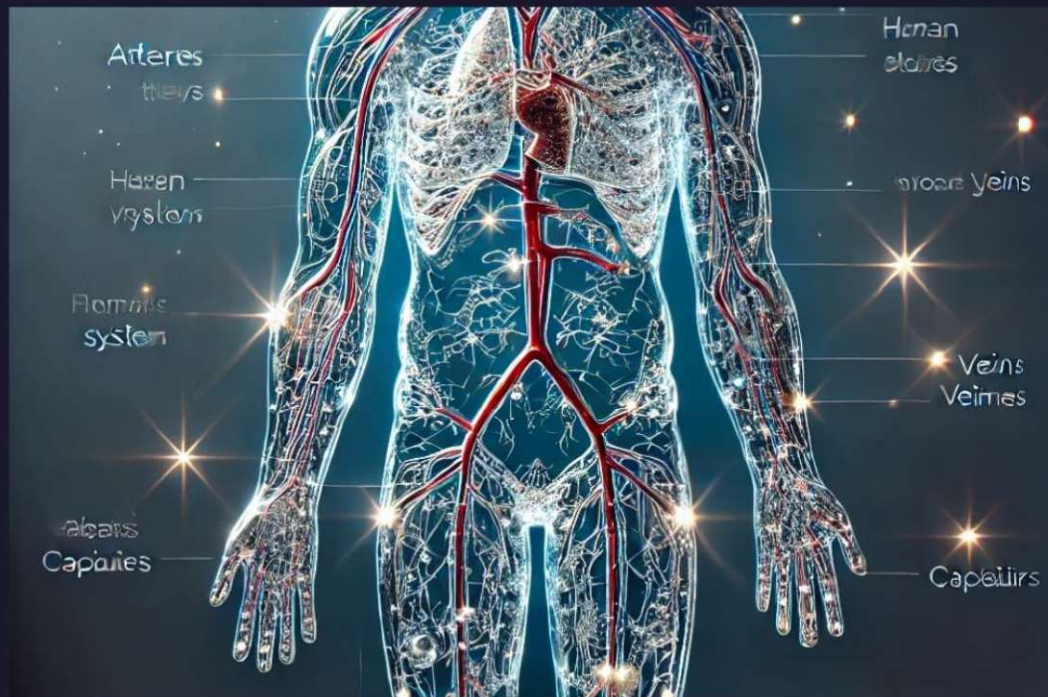
**“k-means
clustering
(unsupervised
learning)
”**

Table 1. Chronic conditions included in definition of multimorbidity (N = 16).

Allergies
Hypertension
High cholesterol
Diabetes (type 1 and type 2)
Heart disease
Stroke
Back pain
Joint disease
Osteoarthritis
Osteoporosis
Chronic obstructive pulmonary disease
Cancer
Dementia
Anxiety
Long-term use of antidepressants
Schizophrenia

<https://doi.org/10.1371/journal.pone.0214183.t001>

Cluster characteristics



- Cluster 1: **The Allergy cluster.** 100% patients has allergy, 42% hypercholesterolemia, 29% COPD 20%, depression, 20%, anxiety, cluster size 63.726, gender women 66%, mean age 57 years, mean number of conditions 2,6.
- Cluster 2: **The Hypercholesterolemia cluster.** 100% patients has hypercholesterolemia, 85% hypertension, none has diabetes and Chronic Heart Disease (CHD) 67.898 cluster size, 45% women, mean number of conditions 2,8.
- Cluster 3: **The Hypertension cluster.** 69% has hypertension, 28% osteoporosis, 26% depression, 24% COPD, 25% gout, 21% back symptoms, 20% cancer, cluster size 63.965, 67% women, mean number of conditions 2,4.
- Cluster 4: **The Diabetes Cluster.** 100% patients has diabetes, 82% hypertension, 72% hypercholesterolemia, none has CHC, cluster size 53.725, 48% women, mean number of conditions 3,3.
- Cluster 5: **The CHD cluster.** 100% patients has CHD, 75% hypertension, 75% hypercholesterolemia, 29 % diabetes, 52.443 cluster size, 44% women, mean number of conditions 3,9.





“Frequentist framework with MLE/EM”

Zero-inflated models:

- Use of hospitalizations and bed days.
- Hospitalizations and bed days were adjusted for emergency visits, out-patient visits, GP visits, out-of-hours GP visits, yearly controls in general practice, private specialist visits, number of conditions, age, gender, cohabitation status, education attainment, and employment status.

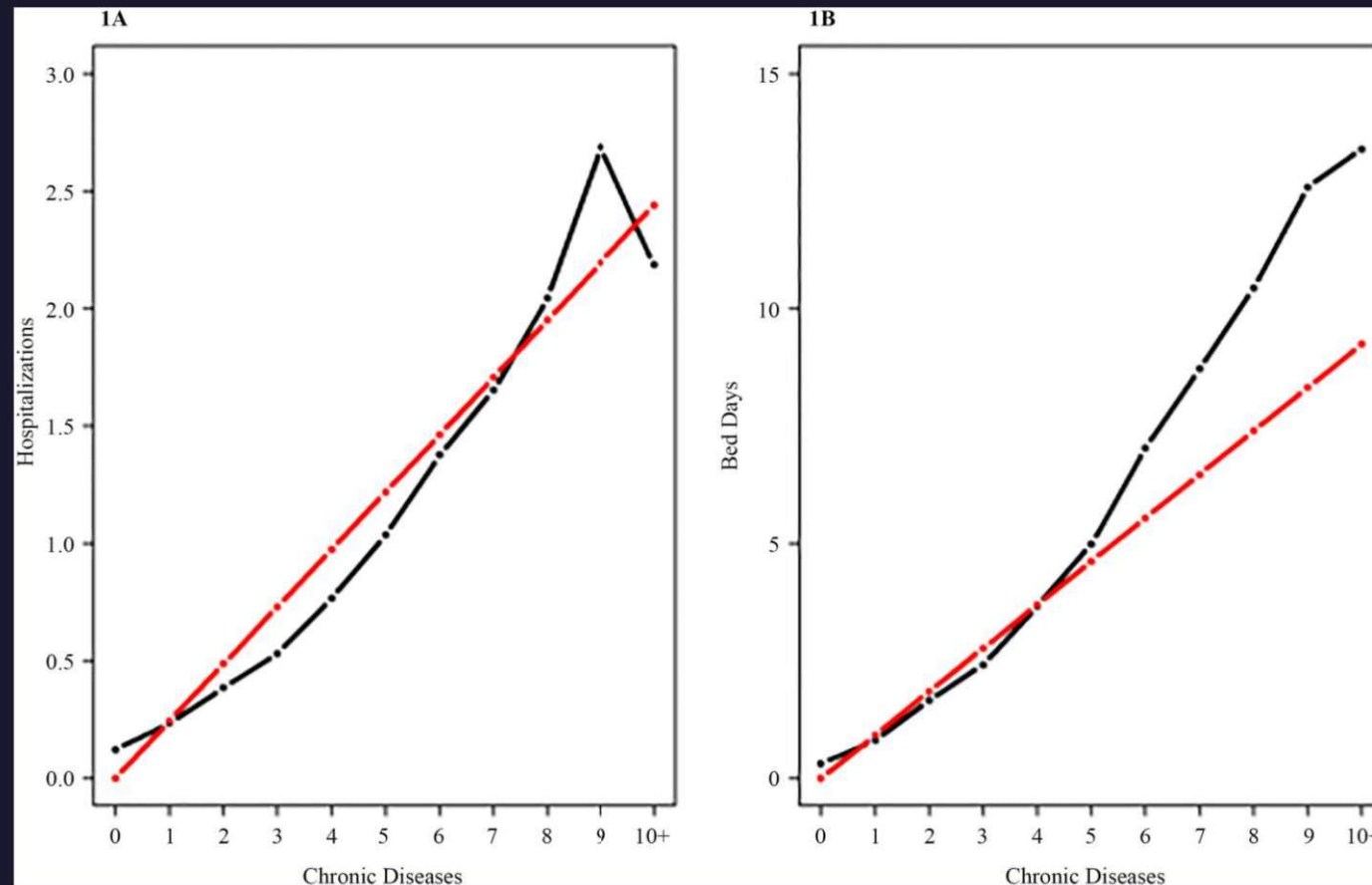


Fig 1. Multimorbidity, healthcare utilization and socioeconomic status: A register-based study in Denmark
Relationship between numbers of chronic conditions and hospitalizations **(1A)** and between numbers of chronic conditions and bed days **(1B)**.

(1A) The black line indicates the observed number of hospitalizations by the number of chronic conditions. The red line indicates the reference regression line that shows the mean number of hospitalizations multiplied by the number of chronic conditions. **(1B)** The black line indicates the observed number of bed days by number of chronic conditions. The red line indicates the reference regression line that shows the mean number of bed days multiplied by the numbers of chronic conditions.

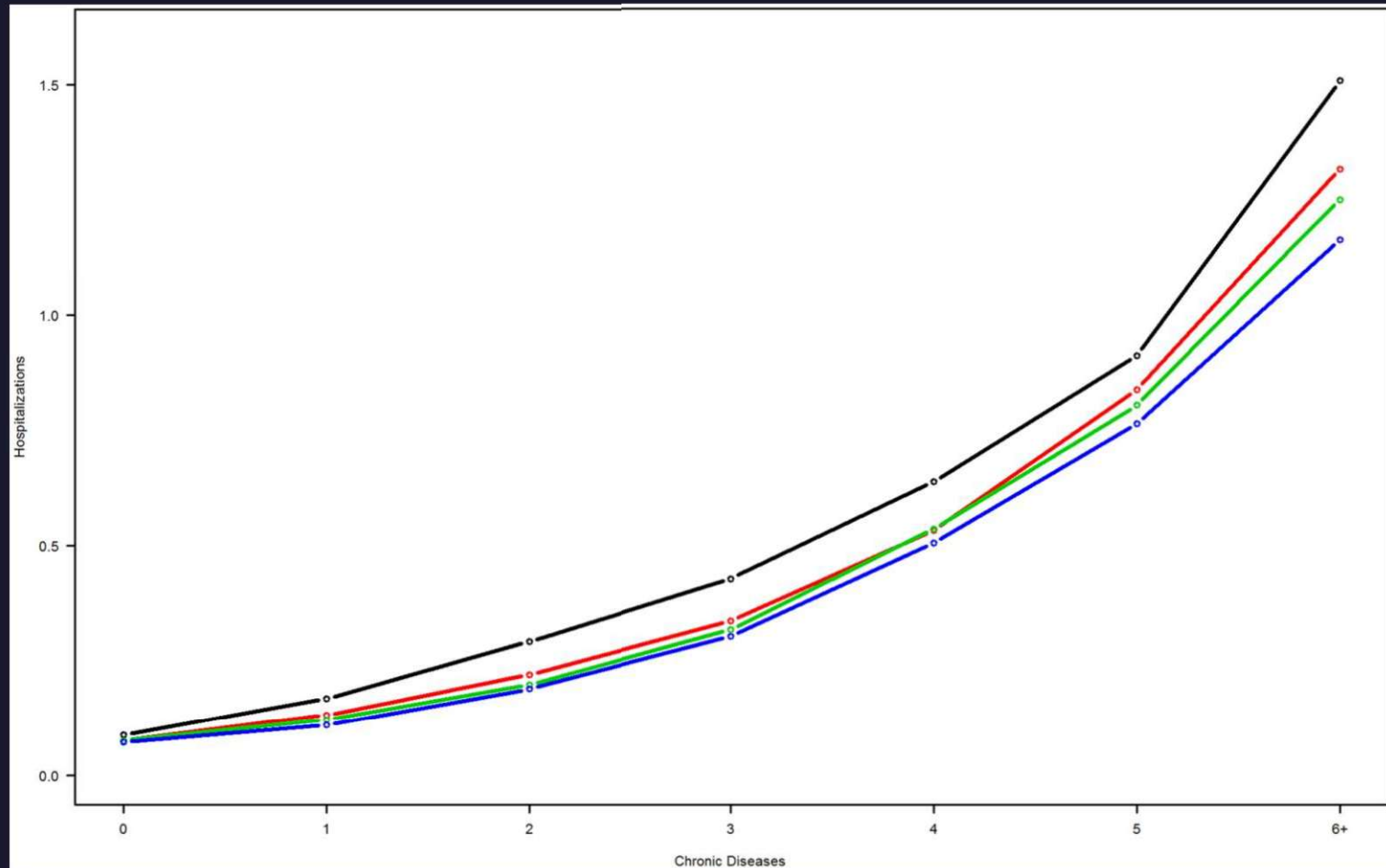


Fig 2. Number of chronic conditions associated with the modeled rate of hospitalizations by educational attainment levels. Black line, no education; red line, short education; green line, medium education; blue line, long education.



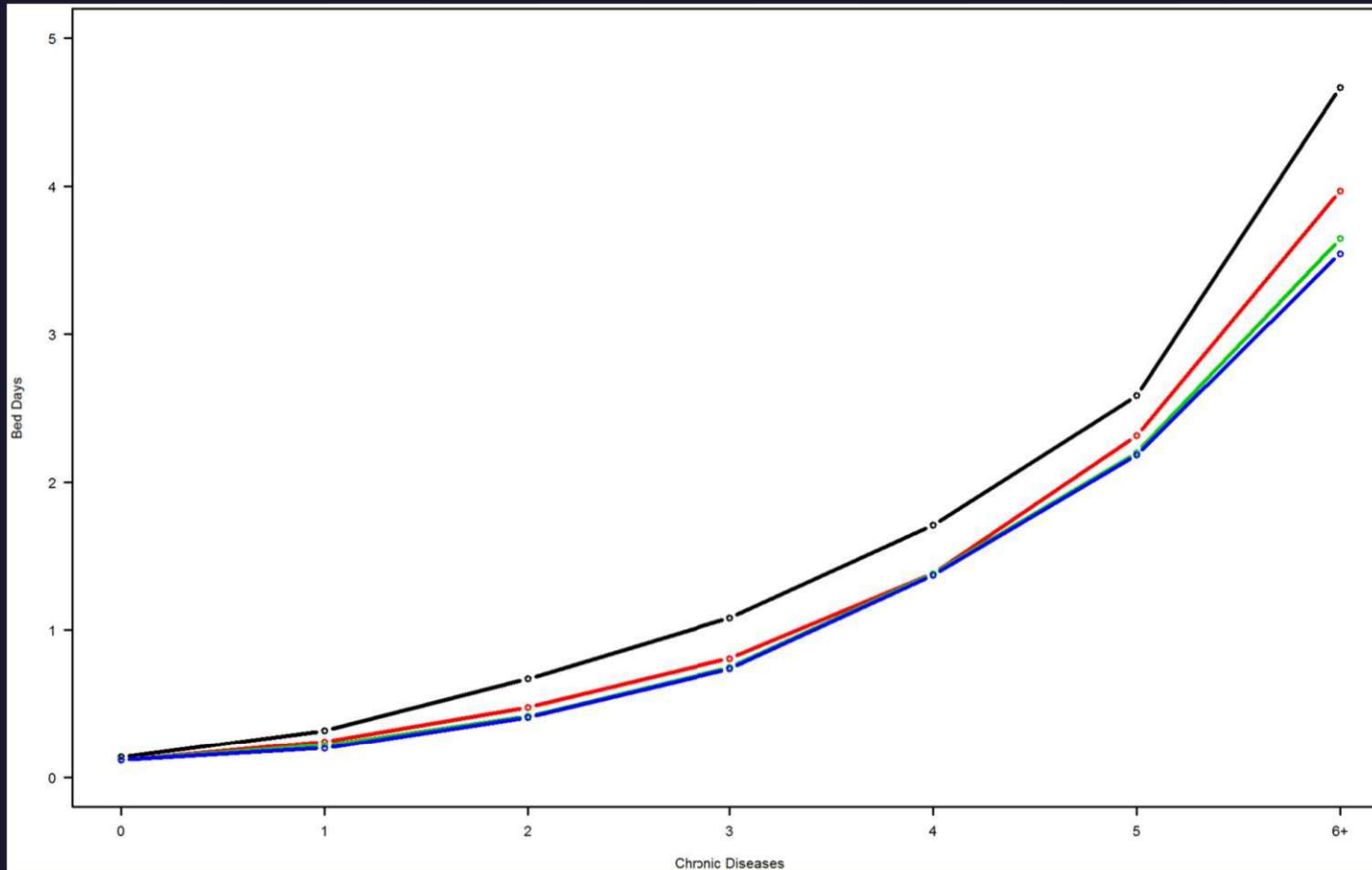


Fig 3. Number of chronic conditions associated with the modeled rate of **bed days** by educational attainment. Black line, no education; red line, short education; green line, medium education; blue line, long education.

ALGORITHM TO BEDSIDE

All models are wrong. Some are Useful!

